Advanced Algorithms 2026 Deadline: 18/10/2026, 23:59 (Zurich time)

Graded Homework 1

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Note 1: *Solutions must be typeset in LaTeX and uploaded via moodle by 23:59 Zurich time on October 18, 2026 (please use a separate pdf file for each problem). Late submissions will not be graded. If you would like to add a drawing in your solution, simply include a picture of a hand-drawn figure in your LaTeX. You can submit only one file for each problem, and the drawing should be incorporated in the rest of your explanation.*

Note 2: *This is a theory course, and any claim should be substantiated with a proof. When asked to give an algorithm, unless the problem statement states otherwise, the algorithm has to have a polynomial running time. You are required to prove the claimed properties of any algorithm you present.*

Note 3: *Please submit the Declaration of Originality along with the graded homework. This is a declaration that you have written your own independent solutions and have not taken help from other students or any AI.*

1 Placing fire trucks (15 Points)

Consider a canton with n villages, where village $v \in [n]$ has a population of $p_v \geq 0$. You want to pair fire trucks with firefighting teams to cover as many people as possible. However, not every team was trained for every possible truck, and there are only m possible pairs of team and truck. Moreover, teams perform differently depending on the truck with which they work; therefore, for every pair $i \in [m]$ we specify the set $V_i \subseteq [n]$ of villages that are *covered*, i.e. the set of villages that the firefighting team can reach in 15 minutes using that truck. Naturally, a *feasible* collection $S \subseteq [m]$ of pairs assigns each team to at most one truck and each truck to at most one team. The total population covered by S is then

$$f(S) = \sum_{v \in \bigcup_{i \in S} V_i} p_v.$$

Design a polynomial-time greedy algorithm that finds a feasible collection of pairs covering at least OPT/3 people, where OPT is the largest population that any feasible collection can cover.

2 Water Quality Testing (25 Points)

You work for a water treatment company that needs to determine which of t possible contaminants are present in a water sample before choosing an appropriate treatment. There are n laboratories, each offering m testing devices. The device $j \in [m]$ offered by the laboratory $i \in [n]$ tests the presence of a set of contaminants $S_{ij} \subseteq [t]$ and can be purchased at a cost $c_{ij} \geq 0$. Additionally, each laboratory charges a delivery fee $d_i \geq 0$, paid once if you order any of its devices, regardless of how many you order.

Your goal is to order a set of devices so that every possible contaminant can be tested while minimizing the total equipment and delivery costs. Assume that every contaminant can be tested with at least one available device.

1. Show that this natural LP has an unbounded integrality gap, even with one laboratory and

one contaminant. (3 Points)

$$\begin{aligned} \text{minimize} \quad & \sum_{i\in[n]} d_i y_i + \sum_{i\in[n]} \sum_{j\in[m]} c_{ij} x_{ij} & \\ \text{subject to} \quad & \sum_{i\in[n]} \sum_{j\in[m]:k\in S_{ij}} x_{ij} \geq 1 & \forall k \in [t] \\ & x_{ij} \leq y_i & \forall i \in [n], j \in [m] \\ & 0 \leq y_i, x_{ij} \leq 1 & \forall i \in [n], j \in [m] \end{aligned}$$

2. Strengthen the LP by adding variables $z_{ik} \geq 0$ and the constraints

$$\begin{aligned} z_{ik} &\leq y_i & \forall i \in [n], k \in [t] \\ z_{ik} &\leq \sum_{j\in[m]:k\in S_{ij}} x_{ij} & \forall i \in [n], k \in [t] \\ 1 &\leq \sum_{i\in[n]} z_{ik} & \forall k \in [t] \end{aligned}$$

   Show that the resulting LP is a valid relaxation. (2 Points)

3. Use the LP to obtain a randomized procedure that selects devices with expected cost at most OPT, where OPT is the optimal LP value, and such that each contaminant is tested with constant probability. (15 Points)

4. Obtain an $O(\log n)$-approximation with probability at least $1 - \delta$ for constant $\delta \in (0, 1)$. (5 Points)

Note. In this problem, the number of contaminants t, the number of laboratories n and the number of devices offered per laboratory m are given as input.

3 Scheduling jobs with quadratic load cost (30 points)

Consider n jobs with nonnegative processing times $p_1, \ldots, p_n$. The goal is to find a schedule of the n jobs on m identical parallel machines that minimizes the sum of squared machine loads, i.e. minimize $\sum_{k=1}^{m} {L_k}^2$, where L_k is the total processing time of jobs assigned to the k-th machine (zero for an empty machine).

1. Suppose that, for some fixed constant C, there are at most C distinct processing times. Give a polynomial-time algorithm that solves the problem exactly in this special case. (5 Points)

2. Give a PTAS for the problem in the general case. (25 Points)
   You may assume the result of Question 1, even if you did not complete that question.

Note. In this problem, the number of jobs n and machines m is given as input.

Note. Recall that a PTAS for a minimization problem is an algorithm that for any fixed constant $\varepsilon > 0$, in polynomial time (where the degree of the polynomial can depend arbitrarily on ε), finds a solution of value at most $(1 + \varepsilon)\text{OPT}$.

4 Currency exchange (30 points)

You have one euro that you must exchange into Swiss francs during the next $T \geq 2$ days. You know T in advance. On day t, you observe the exchange rate $x_t \in [1, 2^k]$, where $k \geq 1$ is a known integer. You can decide whether to exchange your euro or wait for a later day. You must exchange the entire euro in one transaction, no later than day T. No reverse exchanges are allowed.
Your goal is to maximize the number of Swiss francs received. The offline optimum is $\text{OPT} = \max_{t \in [T]} x_t$.

1. Give a deterministic $2^{k/2}$-competitive algorithm and prove that no deterministic algorithm has a better competitive ratio. (10 Points)
2. Give a randomized $2k$-competitive algorithm. (10 Points)
3. Prove that every randomized online algorithm has competitive ratio at least $k/3$ against an oblivious adversary. (10 Points)

Note. An algorithm is c-competitive if its revenue is at least OPT/c. A randomized algorithm $\mathcal{A}$ is c-competitive if its expected revenue, over its random choices, is at least OPT/c for every rate sequence fixed independently of those choices.